

Question: 1

Given the network 192.168.10.0/24, divide it into 4 equal subnets and write down the network address, first usable IP, last usable IP, and broadcast address for each subnet?

Answer: 1. Subnet 1

Network Address: 192.168.10.0

First Usable IP: 192.168.10.1

Last Usable IP: 192.168.10.62

Broadcast Address: 192.168.10.63

2. Subnet 2

Network Address: 192.168.10.64

First Usable IP: 192.168.10.65

Last Usable IP: 192.168.10.126

Broadcast Address: 192.168.10.127

3. Subnet 3

Network Address: 192.168.10.128

First Usable IP: 192.168.10.129

Last Usable IP: 192.168.10.190

Broadcast Address: 192.168.10.191

4. Subnet 4

Network Address: 192.168.10.192

First Usable IP: 192.168.10.193

Last Usable IP: 192.168.10.254

Broadcast Address: 192.168.10.255

Question: 2

You are setting up a Zomato-style food delivery office with 3 departments: Customer Support (50 devices), Delivery Management (30 devices), and Marketing (20 devices). Design a subnetting plan using the 10.0.0.0/24 network so that each department has its own subnet, maximizing available addresses and minimizing waste. List the subnet mask and IP range for each department?

Answer: 1. Customer Support (50 devices)

Subnet Mask: 255.255.255.192 (/26 - Allocates a block of 64 IPs, giving 62 usable addresses).

Subnet Address: 10.0.0.0/26

Usable IP Range: 10.0.0.1 to 10.0.0.62

Broadcast Address: 10.0.0.63

2. Delivery Management (30 devices)

Subnet Mask: 255.255.255.224 (/27 - Allocates a block of 32 IPs, giving 30 usable addresses).

Subnet Address: 10.0.0.64/27

Usable IP Range: 10.0.0.65 to 10.0.0.94

Broadcast Address: 10.0.0.95

3. Marketing (20 devices)

Subnet Mask: 255.255.255.224 (/27 - Allocates a block of 32 IPs, giving 30 usable addresses).

Subnet Address: 10.0.0.96/27

Usable IP Range: 10.0.0.97 to 10.0.0.126

Broadcast Address: 10.0.0.127

### Question: 3

Troubleshoot the following scenario: Two teams in your Flipkart warehouse are accidentally assigned overlapping subnets (Team A: 172.16.10.0/25, Team B: 172.16.10.64/26). Identify the overlapping IP range and suggest a corrected subnet plan so both teams have unique IP ranges.  
  
**Hint:** Draw the address ranges for both subnets to spot the overlap?

#### Answer: 1. Identifying the Overlapping IP Range

Team A (172.16.10.0/25): The block spans from 172.16.10.0 to 172.16.10.127.

Team B (172.16.10.64/26): The block spans from 172.16.10.64 to 172.16.10.127.

The Overlap: 172.16.10.64 to 172.16.10.127. Team B's entire network range is completely engulfed inside Team A's larger subnet, which will cause IP conflicts and routing failures.

#### 2. Corrected Subnet Plan (Option A - Keeping Current Capacities)

To resolve the conflict with minimal disruption, we can shift Team B immediately out of Team A's space into the next available block:

Team A Subnet: 172.16.10.0/25

Subnet Mask: 255.255.255.128

Usable IP Range: 172.16.10.1 to 172.16.10.126

Broadcast Address: 172.16.10.127

Team B Subnet (Shifted): 172.16.10.128/26

Subnet Mask: 255.255.255.192

Usable IP Range: 172.16.10.129 to 172.16.10.190

Broadcast Address: 172.16.10.191

#### 3. Corrected Subnet Plan (Option B - Split Equal Blocks)

If both teams require an equal division of the space (up to 62 usable devices each), split the base range down the middle:

Team A Subnet: 172.16.10.0/26

Usable IP Range: 172.16.10.1 to 172.16.10.62

Team B Subnet: 172.16.10.64/26

Usable IP Range: 172.16.10.65 to 172.16.10.126

Question: 4

You are managing a growing IPL Fantasy League app backend and notice that your 192.168.1.0/26 subnet is running out of IPs. Propose a solution to handle address exhaustion so new servers can be added without disrupting existing

users.  
  
**Hint:** Consider subnet resizing or migrating to a larger subnet?

Answer: 1. Solution A: Expand the Subnet Mask (Vertical Resizing)

Action: Change the subnet mask of your current network block from /26 (255.255.255.192) to /25 (255.255.255.128) or /24 (255.255.255.0).

New Capacity: Upgrading to a /25 expands the available pool from 62 to 126 usable IP addresses, while a /24 gives 254 usable IPs.

Zero Disruption Approach: Because you are only modifying the mask to include additional bits on the right, your existing server IPs (which lie in the .1 to .62 range) remain completely valid within the new, larger subnet boundary. You only need to update the subnet mask configuration on the core router and DHCP server.

2. Solution B: Implement a Secondary Subnet (Multi-Netting)

Action: Keep the current 192.168.1.0/26 block exactly as it is for existing live servers, and allocate a brand-new, adjacent block—such as 192.168.1.64/26 or 192.168.2.0/24—for the new server expansion.

Implementation: Configure your router or Layer 3 switch interface to support IP Secondary addresses or create a new VLAN for the second subnet.

Why It Is Safe: Existing production servers do not require any configuration changes or reboots. The router bridges the traffic smoothly between the old servers and the new servers.

### 3. Solution C: Network Address Translation (NAT) Pool Expansion

Action: If the exhaustion is happening because of public-facing user sessions hitting your backend, implement a Dynamic NAT or PAT (Port Address Translation) pool expansion at your gateway firewall.

Effect: This allows multiple external fantasy app users to share a single backend local IP address using distinct port numbers, instantly reducing the internal address footprint.

### Question: 5

After completing your subnetting speed round and mock exam, review your answers and identify 3 types of mistakes you made (e.g., miscalculating subnet masks, overlapping ranges, wrong broadcast address). For each, write how you would avoid the mistake next time?

Answer: 1. Miscalculating Subnet Masks (e.g., Confusing /26 with /27)

The Mistake: Accidentally writing 255.255.255.192 instead of 255.255.255.224 because of rushing through the binary bit additions.

How to Avoid It Next Time: Memorize or quickly write down a "Magic Number" cheat sheet at the top of your rough sheet as soon as the exam starts:

/25 = 128 (Block size: 128)

/26 = 64 (Block size: 64)

/27 = 32 (Block size: 32)

/28 = 240 (Block size: 16)

## 2. Off-by-One Broadcast Addresses

**The Mistake:** Forgetting that the broadcast address must be an odd number matching the absolute end of the block, resulting in calculations like .64 instead of .63.

**How to Avoid It Next Time:** Always calculate the Next Network Address first. Then, simply subtract 1 from it to get your current subnet's broadcast address.

**Example:** If Subnet 2 starts at 10.0.0.64, Subnet 1's broadcast address must be exactly  $64 - 1 = .63$ .

## 3. Overlapping Subnet Ranges in VLSM

**The Mistake:** Accidentally starting a smaller subnet inside the boundary of a larger, previous subnet because the block boundaries were not tracked sequentially.

**How to Avoid It Next Time:** Sort your host requirements from largest to smallest before doing any math. Draw a straight network number line on paper to visually block out the addresses as you assign them, making sure no two colored blocks overlap.